

Leonardo Bertazzolo

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GitHub: leorooop | **Portfolio:** <https://leorooop.github.io>

I navigate my life driven by pure, lifelong passion for computers, robotics, and the limitless worlds of tech, fantasy, and video games. I bring a unique blend of traits to the tech world: the deep strategic mindset of a competitive Pokémon player, the creative energy of a hard rock soul guitarist, the rigorous discipline of a Ju-Jitsu black belt, and the battle-tested resilience of a Spartan racer.

AI Engineer and Tech Leader with over 10 years of software engineering experience spanning backend development, embedded systems, and enterprise architecture. Proven track record of transitioning from full-stack and Java development to leading major Artificial Intelligence projects.

Expert in designing architecture of scalable microservices ecosystems integrating any kind of technology following most updated design patterns and best practices.

Team leader focused on delivering cost-effective and high-performance automation solutions for challenging client environments.

Technical Skills

- **Architecture & Cloud:** AWS, GCP, Microservices Architecture, Event-Driven Architecture (Apache Kafka, Google Pub/Sub)
- **AI & Data Engineering:** Custom RAG Pipelines, MLflow (MLOps, registries, tracking), LangChain, Vector Databases (Redis, pgvector), Graph Databases (Neo4j, Amazon Neptune)
- **Programming Languages:** Python, Java, C++, C, SQL, JavaScript
- **Frameworks & Libraries:** Spring Framework (Spring State Machine), FastAPI, Pydantic, SQLAlchemy, Alembic, Celery, Qt, QML, ObsPy
- **Testing & Quality Assurance:** Pytest, JUnit, Mockito, Testcontainers, Sonarqube
- **DevOps & Observability:** Docker, Mirantis, Freelen, Splunk, Yocto Project, Jenkins, Kubernetes

Work Experience

AI Engineer / Tech Leader / Solution Architect | Globant

02/2024 - Current | Italy

- **Architectural Leadership:** Take full responsibility for technical direction and solution design of complex microservices ecosystems integrating advanced AI technologies.
- **AI Implementation:** Lead multiple highly successful AI Proofs of Concept, key technical reference for AI integration within process automation.

- **Custom Data Pipelines:** Engineering cost-effective, custom-built RAG data pipelines from scratch to optimize performance over expensive off-the-shelf solutions.
- **Team Management:** Lead international development team (20 people) in a high-pressure client environment as a Java Developer as well as lead python teams (6 people) on projects from scratch to production.
- **Quality & Monitoring:** Enforced rigorous testing standards using Pytest, JUnit, and Testcontainers, while managing application health via Mirantis, Freelens, and Splunk.

Software Engineer | Lunitek Srl

10/2019 - 02/2024 | Italy

- **Team Coordination:** Successfully managed and coordinated a 4 people development team for a specialized research project funded by Tuscany Region.
- **Software Development:** Engineered graphical and command-line data analysis applications using Python and the ObsPy framework to provide customized seismic solutions to clients.
- **Firmware Engineering:** Programmed microcontrollers in C and C++ for integrated seismic and structural monitoring systems, handling power management and ADC/GPS data acquisition.
- **Hardware Initiatives:** Developed standalone electronics projects, including an intelligent solar-panel battery charger for boats and a GPS base station.

Software Engineer | Callegari Srl

11/2018 - 10/2019 | Italy

- **Application Design:** Developed robust application software for an ARM-based touch screen device using C++, Qt, and QML for the UI and core logic.
- **Embedded Linux:** Built and maintained a custom Linux operating system tailored to the device hardware using the Yocto Project.
- **Prototyping:** Utilized OpenCV and GStreamer for image processing and prototyped physical device enclosures using 3D printing.

Software / Automation Engineer | Rimar Srl

05/2017 - 11/2018 | Italy

- **Industrial Automation:** Configured hardware and developed software for Siemens and Allen Bradley PLCs and industrial Fanuc robots.
- **Commissioning:** Executed testing, fine-tuning, and commissioning of automation lines directly at client sites.

Earlier Experience

- Full Stack Software Developer | Zucchetti Spa (10/2016 - 05/2017): Contributed to a Java/JavaScript web-based management system for workplace safety, utilizing SQL Server, PostgreSQL, and Oracle DB.

Education

Bachelor's Degree in Computer, Electronic and Telecommunications Engineering (EQF 6)

Università degli studi di Parma | 01/09/2014 - 17/03/2017 | Parma, Italy

- **Thesis:** Design and implementation of the base joint for an anthropomorphic manipulator.
- **Core Focus:** Computer science, telecommunications, control theory, software/hardware design.

Language Skills

- **Italian:** Native
- **English:** Professional
- **French:** Beginner
- **Spanish:** Beginner

Interests

- Pokémon VGC competitor: winner in international tournaments.
- Guitar player (music academy).
- Black belt in Japanese Ju-Jitsu with over 15 years practice.
- Spartan Race: multi trifecta finisher.